

LLM Cross-Sectional Rankings as Implicit Factor Models

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LLM CROSS-SECTIONAL RANKINGS AS IMPLICIT FACTOR MODELS

From Point-in-Time Data to Portfolios, Factors, and Diagnostics

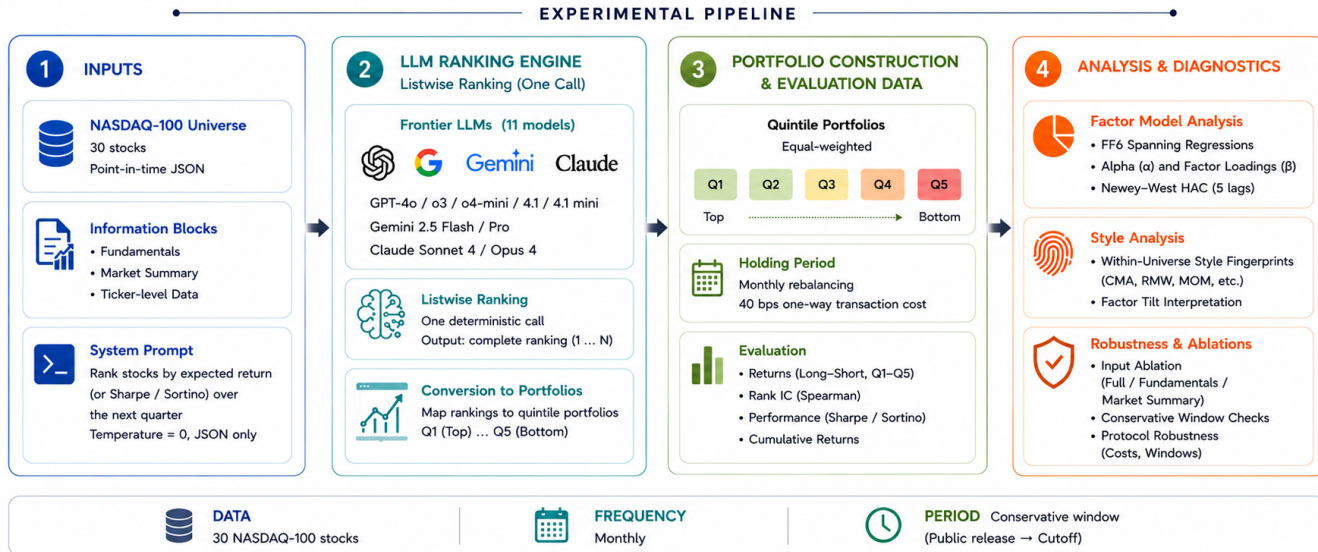


Figure 1: Experimental pipeline for deterministic listwise LLM ranking and portfolio, factor, ablation, and cost diagnostics.

ABSTRACT

Can frontier LLMs be evaluated as factor-like cross-sectional rankers rather than as point forecasters? We study 11 included LLMs from OpenAI, Google, and Anthropic on 30 NASDAQ-100 constituents using a controlled listwise ranking protocol, with expected return as the primary objective and Sharpe/Sortino prompts as robustness checks. Each monthly ranking is converted into quintile portfolios and evaluated with rank IC, FF6 spanning regressions, within-universe style fingerprints, common-window checks, and transaction costs. We find that (i) model choice materially changes the induced factor profile: in the conservative window, o3 reaches a long-short Sharpe of 5.54 and IC of 0.59, while GPT-4o variants produce weak or negative spreads; (ii) the strongest LLM portfolios in this benchmark carry large FF6 alphas with positive momentum and negative value/profitability/investment loadings, consistent with a growth-momentum implicit prior; and (iii) input ablations show that market summaries reproduce most of the full-input factor profile, whereas fundamentals-only prompts weaken both alpha and style coherence. The evidence is a controlled model-comparison benchmark rather than a production trading claim: choosing an LLM changes the implicit factor exposure being tested. An anonymous code and artifact repository is available at https://github.com/llm-CS-ranking/llm_CS_ranking.

CCS CONCEPTS

• **Computing methodologies** → **Learning to rank**; *Natural language generation*; • **Applied computing** → **Economics**.

KEYWORDS

large language models, stock ranking, listwise learning to rank, portfolio construction, factor analysis, quantitative finance

1 INTRODUCTION

Large language models have opened new possibilities for financial analysis: unlike traditional quantitative factors, they can synthesize earnings trends, competitive dynamics, and macro context into investment judgments with no task-specific training. The empirical question is *how* to formulate this capability as a reproducible financial task rather than an anecdotal stock-picking exercise. Prior work has relied either on coarse rating classification [30]—too coarse for portfolio construction and typically evaluated on a single model without backtest—or on raw return prediction, which is dominated by noise [15]. We argue that *ranking*—ordering stocks from most to least attractive—is the most natural middle ground: it is strictly more general than classification or regression, sidesteps the magnitude-noise of returns, and directly enables quintile portfolios, long-short strategies, and Spearman rank-IC evaluation. Our factor-model perspective treats each LLM as an implicit cross-sectional prior: given the same point-in-time information set, different models induce different expected-return orderings, style exposures, and residual alphas. The object is therefore a controlled benchmark of model-induced priors, not a claim that a short-window NASDAQ backtest is directly deployable as a production strategy.

We organize the study around four research questions: **RQ1**: do LLM rankings embed distinct factor-like priors beyond classical FF6

premia? **RQ2**: how do factor loadings and alphas change when the LLM sees fundamentals only, market summaries only, or both? **RQ3**: which vendor generation delivers the strongest expected-return signal after transaction costs? **RQ4**: are LLM rankings mechanical rediscoveries of known style sorts, or do they combine market priors with model-specific tilts? Our contributions are designed to make that benchmark auditable: (i) a self-contained evaluation of zero-shot LLM cross-sectional rankings using FF6 spanning tests, within-universe style fingerprints, common-window checks, and transaction costs; (ii) a cross-model benchmark across 11 frontier LLMs, non-LLM factor/ML baselines, and expected-return as the primary objective; (iii) evidence that strong LLMs in this sample deliver large FF6 alphas with a shared growth-momentum tilt but vendor-specific loadings; and (iv) an input ablation showing that market summaries dominate fundamentals in both performance and factor-profile preservation.

2 RELATED WORK

LLM-based equity analysis and portfolio construction. LLMs have been shown to extract financial signals from news and analyst reports [21, 26], financial statements [22], and fundamentals-plus-news data for equity rating [30], though Papasotiriou et al. evaluate only one model and do not backtest investable portfolios. Recent systems move toward full stock selection and portfolio construction: MarketSenseAI and its agentic extension use GPT-4 for holistic selection [10, 11], TradExpert combines expert LLMs for prediction and ranking [6], and FinMem, TradingAgents, FinCon, and MASS introduce memory, debate, manager-analyst, or scaling-simulation agent designs for trading and portfolio management [16, 43, 45, 46]. Complementary threads study LLM-enhanced momentum [1], region-specific prediction [40], exploratory portfolio selection [34, 42], and decision-focused optimization layers [19]. We instead study whether frontier LLMs can act as zero-shot *listwise cross-sectional rankers*—outputting a complete ordering rather than ratings or weights—evaluated via backtesting and factor-analytic diagnostics.

Cross-sectional ranking and learning to rank. Framing stock selection as ranking predates LLMs: traditional factor research sorts on human-defined characteristics [8, 20], and modern ML learns ranking signals from characteristic panels [2, 15]. An equity-specific L2R literature covers sentiment-driven ListNet/RankNet [38], Relational Stock Ranking [12], STHAN-SR hypergraph attention [37], pointwise/pairwise/listwise objectives [31, 47], label design [17], and temporal stock representations [18]. In NLP, LLMs themselves act as zero-shot rankers through RankGPT-style permutation generation and later pairwise, setwise, and listwise prompting [32, 33, 39, 48]. Our contribution is to elicit full listwise cross-sectional rankings *zero-shot* from frontier LLMs without any task-specific financial training.

Factor generation, multi-agent systems, and recommendation. LLMs have also been used to generate predictive factors, mine decay-resistant alphas, and automate quant factor-model R&D [4, 25, 41], or to coordinate financial decisions via multi-agent systems [3, 27, 44]; see Nie et al. [29] for a survey. Our factor analysis (Section 4.3) treats LLM rankings themselves as implicit factors and tests their spanning by classical risk premia. A related but distinct literature on *personalized* asset recommendation [5, 24, 35, 36]

targets user-specific utility; our study instead asks how LLM rankings behave as cross-sectional asset-pricing operators on a fixed investable universe.

3 METHODOLOGY

3.1 Listwise Ranking Formulation

Among pointwise, pairwise, setwise, and listwise L2R paradigms [32, 33, 48], we adopt the *listwise* approach: the LLM receives data for all n stocks simultaneously and returns a single permutation, avoiding the $\binom{n}{2}$ cost of pairwise calls and the coarseness of classification. Rankings preserve the relative information needed for portfolio construction while discarding the magnitude noise that dominates raw return prediction [15], and any classification or regression output can be converted to a ranking but not vice versa.

3.2 Universe and Data

Our universe is the 30 largest NASDAQ-100 constituents by market cap (NVDA, AAPL, MSFT, GOOG, AMZN, META, TSLA, and 23 others)—a fixed, concentrated evaluation panel that is tractable within a single prompt while still supporting cross-sectional ranking. This is a benchmark panel, not a survivorship-free investable universe: fixing the cross-section isolates model-ranking behavior and prompt scalability, while Section 5 treats broader rolling universes as a necessary extension. Each stock-month is described by two JSON blocks: **Fundamentals**—up to 40 S&P Capital IQ items (revenue, net income, EPS, total assets, operating cash flow, etc.) with their four most recent quarterly values; and **Market summary**—nine fields covering 1m/3m returns, 1m/3m annualized volatility, 20d/60d MA gaps, average volume, volume z-score, and drawdown from high. All inputs are point-in-time: only information available before the first day of the prediction month is included.

3.3 Listwise Ranking Protocol

For each month t , prediction mode $m \in \{\text{return, Sharpe, Sortino}\}$, and model M , we (1) build a JSON payload with the fundamentals and market summary for all $n = 30$ stocks as of t , (2) prompt M with a system instruction—“Rank the following stock tickers by their expected [return / Sharpe / Sortino] over the next rebalancing period. Use ascending order (1 = highest). Return JSON only.”—and (3) parse the JSON response into a ranking $r_{i,t} \in \{1, \dots, n\}$. All API calls use temperature = 0 for reproducibility, and malformed JSON responses are discarded rather than repaired by hand. This design makes the benchmark intentionally austere: differences across models arise from the same point-in-time data, the same listwise prompt, and the same deterministic decoding protocol. The user payload is enclosed in <DATA_JSON> tags and contains three keys: fundamentals, market, and notes. The market block includes recent return, volatility, moving-average gap, volume, and drawdown summaries; the notes block records the as-of month and quarter ordering.

3.4 Models

We evaluate 11 frontier LLMs from three vendors—OpenAI (GPT-4o/4o mini/4.1/4.1 mini/4.1 nano, o3, o4-mini), Google (Gemini 2.5 Flash/Pro), and Anthropic (Claude Sonnet 4, Opus 4)—spanning

multiple generations and a wide cost range. For the asset-pricing narrative we highlight one representative model per vendor family—o3, Gemini 2.5 Pro, and Claude Opus 4—while retaining the full 11-model benchmark in the tables. Every prompt contains only point-in-time data available at the as-of date, reducing input-level lookahead; however, for backtesting we do *not* rely solely on the official model-documentation “knowledge cutoff” date. Instead, the admissible boundary is the latest model release or documented update date available at backtest time whenever that is later than the static knowledge cutoff, and we include only full calendar months after that boundary. Expected return is the primary objective throughout; Sharpe and Sortino prompts are reported as robustness checks on risk-adjusted wording.

3.5 Backtest Design and Metrics

Each monthly ranking is converted into equal-weighted quintile portfolios: **Q1** (stocks ranked 1–6), **Q5** (ranked 25–30), and a dollar-neutral long–short **Q1–Q5**. Portfolios are rebalanced on the third trading day of each month (lag = 2 days) with a one-way transaction cost of 40 bps on turnover. We compare against equal-weighted (**UNIV_EW**) and cap-weighted (**UNIV_CAPW**) benchmarks over the same universe. As non-LLM ranker controls, we also evaluate single-characteristic factor sorts (12–1 momentum, 1-month reversal, 60-day low volatility, earnings yield, ROE, and conservative investment), an equal-weighted composite factor ranker, and expanding-window Ridge/Random-Forest return rankers trained only on prior months. All portfolio metrics—CAGR, annualized volatility, Sharpe, Sortino, maximum drawdown (MDD), Calmar, and hit ratio—are computed from daily returns and annualized where applicable to enable comparability across different backtest lengths. To measure ranking quality directly we compute the Spearman Rank IC between the predicted ranking and the realized return ranking over the same monthly holding period, together with its annualized information ratio $ICIR = \bar{IC}/\sigma_{IC} \times \sqrt{12}$. Because parsed evaluation coverage differs across models and recent model windows can be short, we do not treat annualized CAGR as a standalone acceptance criterion. Instead, a ranking is credible only when portfolio separation, rank IC, FF6 spanning tests, same-window controls, and within-vendor generation comparisons point in the same direction. Table 1 summarizes the tested model set, conservative boundary used for filtering, first admissible evaluation month, and approximate per-experiment API cost, making the cost–performance comparisons in RQ3 auditable.

4 RESULTS

4.1 Q1 Performance and Quintile Monotonicity

Figure 2 displays the CAGR of each quintile, the central diagnostic for whether the entire ranking produces cross-sectional separation rather than a single lucky top pick. **o3 is the strongest top-quintile ranker**, reaching a Q1 CAGR of 280.4% (Sharpe = 4.41) in the expected-return mode—far above the benchmarks—with a cleanly monotonic Q1→Q5 pattern. Gemini 2.5 Flash, Gemini 2.5 Pro, and GPT-4.1 show similarly strong top-quintile performance, with expected-return Q1 CAGRs above 240%. In contrast, **GPT-4o mini produces negative Q1 returns** and negative long–short spreads, a signal worse than random in this evaluation window.

Table 1: Model coverage, conservative evaluation starts, and approximate monthly experiment costs.

Model	Vendor	Boundary	Start	Cost
GPT-4o	OpenAI	2024-05	2024-06	\$1.03
GPT-4o mini	OpenAI	2024-07	2024-08	\$1.03
GPT-4.1	OpenAI	2025-04	2025-05	\$1.23
GPT-4.1 mini	OpenAI	2025-04	2025-05	\$0.33
GPT-4.1 nano	OpenAI	2025-04	2025-05	\$0.04
o4-mini	OpenAI	2025-04	2025-05	\$0.55
o3	OpenAI	2025-04	2025-05	\$0.93
Gemini 2.5 Flash	Google	2025-06	2025-07	\$0.14
Gemini 2.5 Pro	Google	2025-06	2025-07	\$0.59
Claude Sonnet 4	Anthropic	2025-05	2025-06	\$1.28
Claude Opus 4	Anthropic	2025-05	2025-06	\$6.38

This first screen answers whether the LLM output is an economically ordered ranking at all before applying factor diagnostics.

4.2 Long–Short Spread

Table 2 reports the market-neutral Q1–Q5 long–short portfolio together with rank IC, with boldface marking the best model within each metric–objective column. o3 is the strongest expected-return spread model in the conservative window, reaching a 466.0% net long–short CAGR, Sharpe 5.54, Sortino 10.70, and IC 0.594 after charging both long- and short-leg turnover. Gemini 2.5 Pro (458.8%, Sharpe 4.99, IC 0.577), Gemini 2.5 Flash (380.7%, Sharpe 4.79, IC 0.513), and GPT-4.1 (367.6%, Sharpe 4.72, IC 0.549) also deliver large positive spreads. GPT-4o and GPT-4o mini produce negative long–short returns in the conservative window, confirming that model choice can flip the sign of the induced cross-sectional prior. Because these are annualized short-window portfolio statistics, the relevant evidence is their agreement with IC, ensemble, same-window, and factor-spanning diagnostics, not the CAGR magnitude alone.

An offline average-rank ensemble check shows that the signal is not purely a single-model artifact. Table 3 averages expected-return ranks across model sets on their common admissible months. The nine-model consensus across GPT-4.1, o-series, Gemini 2.5, and Claude 4 families reaches a 474.2% net long–short CAGR, Sharpe 5.44, and IC 0.549 from 2025-07 to 2025-12, while the OpenAI reasoning consensus reaches Sharpe 5.39.

4.3 Factor Analysis of LLM Rankings

The central asset-pricing question is whether LLM-induced orderings behave like factor-like cross-sectional priors. If rankings simply re-discover known risk premia, their long–short returns should be spanned by standard factor models; if not, they should exhibit distinct loadings and alphas. We address this with two analyses: (i) time-series *spanning regressions* of Q1–Q5 daily returns on established factors [7–9], and (ii) *characteristic-based style analysis* mirroring the classical factor workflow (sort, form long–short, test alpha) but replacing the human sort with the LLM’s ranking. Section 4.4 applies the same logic to fundamentals-only and market-only input ablations.

Spanning test. For each expected-return LLM ranker, we regress gross daily Q1–Q5 returns on FF6 factors [13] with Newey–West HAC SE (5 lags); Table 4 reports annualized alpha, factor loadings, and adjusted R^2 . Three findings stand out: (i) *winning LLMs have*

Quintile Monotonicity Check

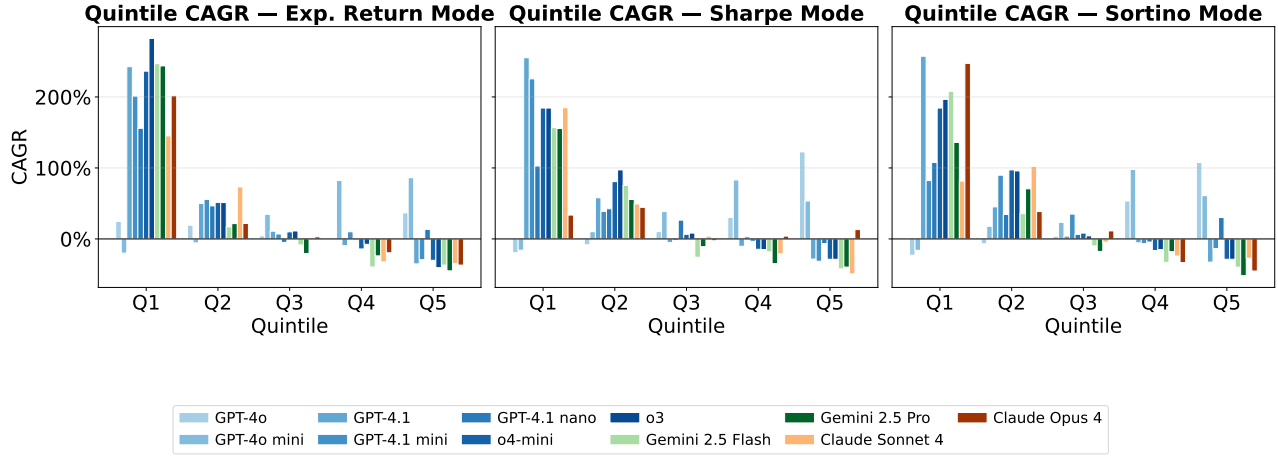


Figure 2: Quintile CAGR by model and prompt objective in each model's conservative evaluation window.

Table 2: Conservative-window Q1–Q5 long–short performance and rank IC, net of 40 bps one-way costs on both legs. Bold indicates the best value in each column.

Ranker	Exp. Return					Sharpe					Sortino				
	CAGR	Shp	Sort	MDD	IC	CAGR	Shp	Sort	MDD	IC	CAGR	Shp	Sort	MDD	IC
GPT-4o	−22.7	−0.62	−0.83	−57.3	0.090	−70.3	−3.31	−3.89	−86.0	−0.264	−69.8	−3.31	−3.92	−85.7	−0.281
GPT-4o mini	−63.1	−4.02	−4.70	−77.0	−0.303	−52.7	−2.92	−3.63	−66.1	−0.241	−55.7	−3.15	−3.86	−69.6	−0.255
GPT-4.1	367.6	4.72	8.37	−11.8	0.549	342.7	4.85	9.29	−11.3	0.540	372.1	4.90	9.30	−10.8	0.510
GPT-4.1 mini	260.3	4.09	7.64	−15.9	0.449	314.0	4.82	8.96	−8.5	0.498	81.2	2.41	4.06	−15.7	0.306
GPT-4.1 nano	100.4	2.82	4.63	−8.8	0.294	88.2	2.75	4.56	−11.6	0.300	42.1	1.80	2.71	−8.9	0.230
o4-mini	333.0	4.65	8.39	−11.4	0.529	258.7	4.48	7.75	−8.5	0.524	258.7	4.48	7.75	−8.5	0.526
o3	466.0	5.54	10.70	−8.7	0.594	258.7	4.48	7.75	−8.5	0.526	274.1	4.61	8.07	−8.5	0.561
Gem. 2.5 Flash	380.7	4.79	8.81	−14.1	0.513	298.5	4.64	8.40	−7.7	0.509	356.7	5.22	10.65	−7.1	0.482
Gem. 2.5 Pro	458.8	4.99	9.15	−11.4	0.577	270.2	4.55	8.37	−6.4	0.511	329.1	5.86	12.38	−4.9	0.569
Cl. Sonnet 4	231.9	3.99	7.24	−9.9	0.534	389.3	5.78	11.80	−6.0	0.588	113.5	2.91	4.77	−14.8	0.539
Cl. Opus 4	312.9	4.56	8.20	−13.0	0.483	−1.6	0.06	0.09	−33.7	0.283	444.9	5.74	11.63	−6.0	0.749

Table 3: Conservative-window average-rank ensemble robustness. Consensus ranks are evaluated as the same net 40 bps Q1–Q5 portfolio.

Consensus	Models	Start	Mo.	LS CAGR	LS Shp	IC
Nine-model consensus	9	2025-07	6	474.2	5.44	0.549
OpenAI 4.1 consensus	3	2025-05	8	306.8	4.49	0.486
OpenAI reasoning consensus	2	2025-05	8	428.9	5.39	0.567
Gemini 2.5 consensus	2	2025-07	6	454.2	4.86	0.542
Claude 4 consensus	2	2025-06	7	317.2	4.57	0.560
Low-cost consensus	3	2025-07	6	379.4	4.89	0.515

large, highly significant alphas—o3 $\alpha = 351\%$ ($t = 5.7$), Gemini 2.5 Pro $\alpha = 340\%$ ($t = 4.7$), and Gemini 2.5 Flash $\alpha = 285\%$ ($t = 4.7$)—so their returns are not mechanically spanned by the standard factor zoo in this benchmark; (ii) adjusted R^2 ranges from 0.08 to 0.52, indicating that standard factors explain part, but not all, of the spread-return variation; (iii) failing models have significantly negative alphas (GPT-4o mini $\alpha = -55\%$, $t = -4.5$), i.e., structurally counter-factorial during the AI-rally period.

Style fingerprint. Because global Fama–French factors are built on the full US cross-section while our universe is 30 mega-cap NASDAQ stocks, they may understate within-universe style tilts. We therefore rank stocks each month on each characteristic (normalized to $[0, 1]$ so that +1 is the factor's traditional “long” side) and average the rank within each LLM's Q1 and Q5. Figure 3 visualizes the Q1–Q5 style fingerprint for the expected-return mode.

A consistent pattern emerges across the strongest models: large positive momentum tilts (+0.23 to +0.32) combined with negative exposures to low-volatility, value, profitability, and conservative-investment—a “growth–momentum chaser” style that worked in the 2024–2025 AI rally and matches their FF6 loadings in Table 4. GPT-4o mini displays the mirror image (negative momentum, positive reversal and volatility), explaining its anti-monotonic returns: it systematically picks defensive, recently-weak stocks in a period when the opposite was rewarded. Within-universe factor-mimicking correlations show the same qualitative pattern at the daily-return level.

Together these results imply that (a) LLM long–short portfolios in this panel are not fully recovered by linear combinations of

Table 4: FF6 spanning tests for conservative-window Q1–Q5 returns. α is annualized percent; stars use Newey–West HAC SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Model	α (%)	$t(\alpha)$	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{MOM}	R^2_{adj}
GPT-4o	−14.2	−0.54	0.34***	0.07	−0.28	−0.03	−0.50*	0.17	0.08
GPT-4o mini	−54.8***	−4.52	−0.06	−0.14	0.21	0.07	0.58***	−0.38***	0.13
GPT-4.1	250.2***	4.73	1.23***	−0.48*	−0.36	−0.92***	−0.23	0.36	0.45
GPT-4.1 mini	225.5***	3.38	0.55**	0.12	0.10	−0.61*	−0.61	0.69**	0.28
GPT-4.1 nano	96.6**	2.39	0.74***	−0.18	−0.11	0.22	−0.39	−0.04	0.10
o4-mini	230.1***	4.61	1.24***	−0.45*	−0.31	−0.69***	−0.42	0.32	0.44
o3	350.8***	5.71	1.15***	−0.47*	−0.29	−0.67***	−0.22	0.25	0.35
Gem. 2.5 Flash	284.7***	4.67	1.33***	−0.32	−0.08	−0.32	−0.54	0.58*	0.48
Gem. 2.5 Pro	340.2***	4.71	1.49***	−0.47	−0.27	−0.48*	−0.40	0.51	0.50
Cl. Sonnet 4	182.9***	5.02	0.95***	−0.46*	−0.09	−0.40	−1.02***	0.45**	0.52
Cl. Opus 4	261.0***	3.33	1.04***	0.01	0.22	0.26	−1.39**	0.12	0.25

classical style factors—consistent with LLM-based factor generation [4]—and (b) each LLM embeds a distinct implicit prior over which characteristics predict returns. Selecting an LLM therefore changes the implicit factor exposure being evaluated.

4.4 Input Ablation and Factor Loadings

The factor-prior interpretation also asks which information channel generates the implicit ranking profile. We re-run every model-objective pair with fundamentals only, market summaries only, or both inputs, then repeat the FF6 spanning test for one representative model per vendor in expected-return mode. Table 5 shows that market-only prompts retain most of the measured signal and slightly exceed the full prompt on average in the conservative window, whereas fundamentals-only prompts are much weaker; Panel C links that performance gap to changes in HML, CMA, and momentum loadings for representative models. This supports the interpretation that recent market summaries, rather than sparse accounting panels alone, carry most of the observed ranking signal in this narrow universe.

4.5 Rank IC Analysis

Because Table 2 reports rank IC next to portfolio outcomes, it also checks whether returns arise from genuine cross-sectional ordering rather than a few lucky names. Across the LLM benchmark, **o3 achieves the highest expected-return IC (0.594)**, followed by

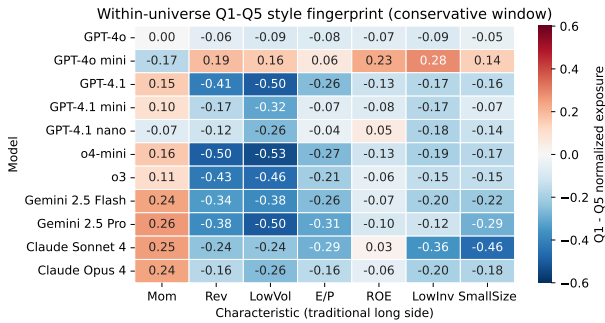
Table 5: Input ablation diagnostics under conservative model windows. CAGR and IC are reported in percent; Panel C stars use Newey–West HAC SE: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Expected-return performance					
Input	Mean LS CAGR	Mean LS Shp	Mean IC	N	
Fund. + Market	256.9	3.23	39.2	11	
Market only	313.8	3.38	48.7	9	
Fund. only	47.6	1.32	11.4	9	
Panel B: Paired tests					
Comparison	Metric	Both	Ablated	Δ	p
Both - Market	LS CAGR	218.2	294.3	−76.1	0.154
Both - Market	LS Shp	2.68	3.21	−0.53	0.478
Both - Market	IC	33.4	43.5	−10.1	0.154
Both - Fund.	LS CAGR	218.2	20.4	197.8	0.021
Both - Fund.	LS Shp	2.68	0.53	2.15	0.096
Both - Fund.	IC	34.1	7.4	26.7	0.053
Panel C: Representative FF6 loadings					
Model	Input	α	β_{HML}	β_{CMA}	β_{MOM}
o3	Fund. + Market	350.8	−0.29	−0.22	0.25
o3	Market only	317.4	−0.19	−0.28	0.46**
o3	Fund. only	84.6	−0.50**	−0.40	0.31
Gem. 2.5 Pro	Fund. + Market	340.2	−0.27	−0.40	0.51
Gem. 2.5 Pro	Market only	414.2	−0.62	0.50	0.47
Gem. 2.5 Pro	Fund. only	76.2	−0.27	−0.69	0.38
Cl. Opus 4	Fund. + Market	261.0	0.22	−1.39**	0.12
Cl. Opus 4	Market only	555.9	−0.20	−0.45	0.13
Cl. Opus 4	Fund. only	−7.2	−0.21	−1.26***	0.18

Gemini 2.5 Pro (0.577), GPT-4.1 (0.549), and Claude Sonnet 4 (0.534). GPT-4.1 mini remains positive after patching malformed months (IC = 0.449), while GPT-4o mini is negative across all modes—its rankings are systematically inversely correlated with realized outcomes.

4.6 Prediction Mode and Cumulative Returns

Across the included models, the **expected-return mode remains the clearest primary benchmark**: it directly targets the holding-period objective and produces the strongest spread for several of the top rankers, while Sharpe/Sortino prompts act more like style-control variants. Sharpe/Sortino prompts win for specific models (e.g., GPT-4.1 reaches its best Q1 CAGR of 257.3% in Sortino mode) but produce more variable results overall, suggesting expected-return prompts remain the clearest primary benchmark. Figure 4 shows the same pattern in cumulative wealth: the stronger post-2025 rankers, including o3, Gemini 2.5, GPT-4.1-family, and Claude

**Figure 3: Within-universe Q1–Q5 style fingerprint in the expected-return mode.** Positive cells tilt toward the characteristic’s traditional long side.

Long–Short (Q1–Q5) Cumulative Wealth

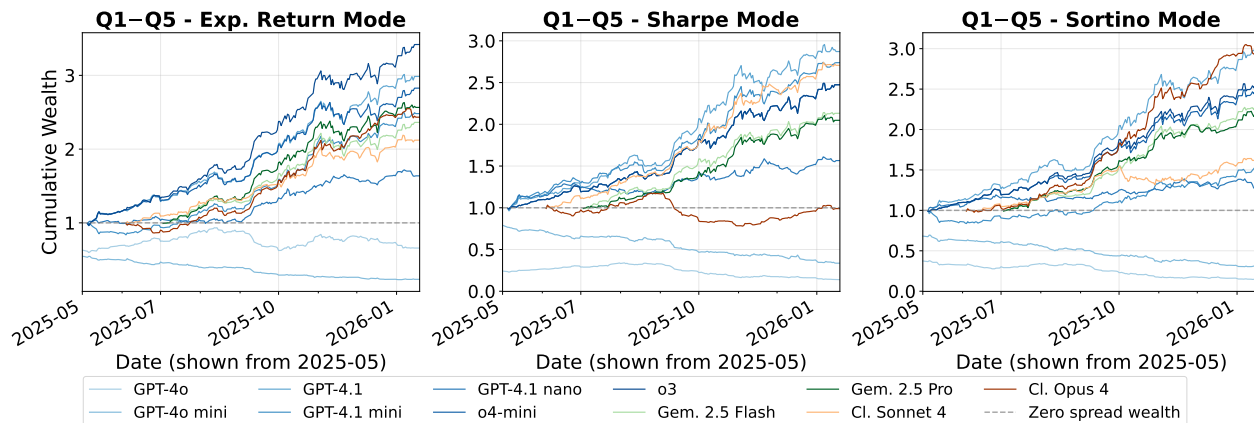


Figure 4: Cumulative Q1–Q5 wealth by model and prompt objective, shown from 2025-05 after compounding each path from its conservative start; the dashed line marks zero spread wealth.

4 models, accumulate positive spreads, while GPT-4o-family models lose money.

4.7 Protocol Checks

Several design choices make the benchmark conservative. First, all LLM and non-LLM rankers are evaluated on the same monthly universe, holding-period definition, 2-day execution lag, and 40 bps one-way transaction cost charged to both long- and short-leg turnover. The performance rows in Table 2 therefore share the same market data, universe, rebalance schedule, and transaction-cost accounting, so differences reflect the induced rankings rather than different portfolio mechanics. Second, the model table fixes the information boundary induced by knowledge cutoffs. Because newer models have shorter admissible post-cutoff windows, we avoid interpreting raw CAGR alone and instead emphasize within-vendor generation comparisons, rank IC, same-window non-LLM controls, and FF6 alphas. The conservative boundary prevents newer preview models from borrowing pre-release 2025 months and makes the negative GPT-4o-family results directly visible. After patching malformed non-deprecated OpenAI outputs, all included model-mode panels have at least 20 ranked names in each evaluated month.

Third, the listwise protocol uses one deterministic call per model-month-objective rather than prompt search, voting, or hand repair. Malformed JSON is discarded rather than manually corrected, and temperature is fixed at zero. This keeps the reported differences attributable to the ranker induced by a frontier model under a common prompt, not to downstream engineering. It also makes the benchmark a useful lower bound for future agentic systems: an agentic ranker, pairwise tournament, or ensemble should be compared against these single-call listwise results under the same information set and cost accounting.

5 DISCUSSION

Scope of the evidence. The results should be read as a stress test of model-induced rankings under a fixed information set. The fixed 30-stock NASDAQ panel, short post-release windows, simplified shorting assumptions, and possible hidden memorization channels

limit direct claims about deployable trading performance. Their value is instead diagnostic: the same protocol reveals which LLMs induce monotone rankings, positive IC, distinct style loadings, and residual alphas under conservative timing.

Model generation effect. The most striking finding is the gap between generations and vendors: within OpenAI, o3’s expected-return IC of 0.594 and GPT-4.1’s 0.549 greatly exceed GPT-4o’s 0.090 and GPT-4o mini’s -0.303 ; within Google, Gemini 2.5 Pro and Flash both deliver strong positive ICs despite short conservative windows. The pattern is directionally consistent with model capability: newer reasoning models and larger frontier models tend to produce stronger rankings, although price is not a sufficient statistic because smaller models such as Gemini 2.5 Flash and GPT-4.1 mini remain competitive.

Ranking vs. classification. Full rankings expose signal structures that classification would hide: GPT-4o mini’s *anti-monotonic* quintile returns would be invisible under a 5-category Buy/Hold/Sell scheme, and o3’s market-neutral long–short Sharpe of 5.54 is only attainable when rankings separate both the top and bottom of the cross-section. In our setting, the bottom quintile is as informative as the top: strong models avoid the same stocks that weak models over-rank, converting a long-only selection signal into an investable spread. Buy/Hold/Sell labels or top- k hit rates would therefore understate both positive and negative model skill.

Cost-effectiveness and reproducibility. Gemini 2.5 Flash, GPT-4.1 mini, and GPT-4.1 nano offer strong cost-performance trade-offs among included models: all are inexpensive relative to larger frontier or reasoning models yet deliver positive spreads after conservative-window filtering. High-cost models such as Claude Opus 4 are less attractive on a spread-return basis. The relevant unit is not leaderboard capability but ranking quality per dollar under a fixed information set.

Cost-Effectiveness Analysis

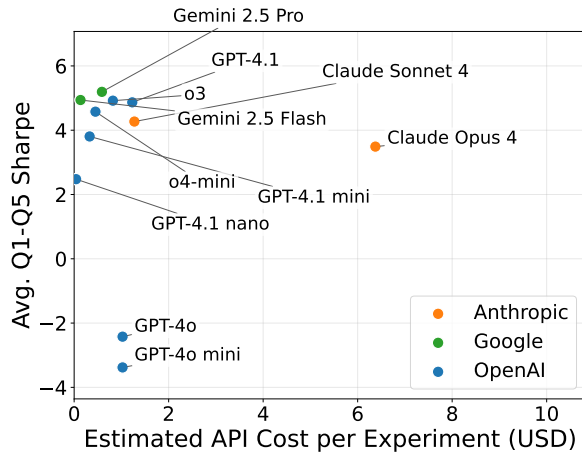


Figure 5: Cost versus average Q1–Q5 Sharpe for tested LLM rankers.

Cost should be reported at the experiment level rather than only per token because pairwise or agentic variants can multiply API usage quickly.

Risk-adjusted prompts as implicit risk control. Prompting for Sharpe/Sortino acts as implicit risk control even without weights or an optimizer: the risk-adjusted objectives often reduce drawdowns or change which names enter Q1 and Q5. The factor analysis (Section 4.3) confirms a genuine style shift, moving the implicit style away from a pure “momentum $\times$ growth” profile toward more risk-aware rankings for some models. However, the risk prompts do not uniformly improve ranking quality. They often reduce drawdowns by dampening the high-momentum growth tilt, but they can also dilute or redirect the strongest return signal, as seen in the model-specific differences between expected-return, Sharpe, and Sortino prompts. Prompt choice should therefore be viewed as a style-control knob rather than a free improvement in objective alignment.

IC as diagnostic, vendor risk profiles, and limitations. Mean IC and long–short Sharpe are near-linearly related ($r = 0.90$, $p < 0.001$), making IC a reliable ex-ante diagnostic. Anthropic long–short portfolios show relatively shallow drawdowns in conservative windows, while the GPT-4o family shows much deeper losses and negative ICs. Remaining limitations are important for follow-on work: post-cutoff windows reduce direct leakage from evaluated holding-period returns, but broader LLM memorization, hidden lookahead channels [14, 28], and market reflexivity in historical AI backtests [23] warrant formal testing; backtest windows differ outside the common-window robustness check; and LLM outputs can drift across API versions despite temperature = 0. Some historical model IDs are preview, date-pinned, or currently unavailable for live reruns, so we distinguish cached benchmark evidence from future live-API reproducibility. We therefore interpret the results as a controlled model-comparison benchmark rather than evidence of production tradability. A natural benchmark extension is to add pointwise LLM scoring, pairwise or setwise tournaments, ensembles, and lightweight agentic rankers on the identical information

set. The fixed listwise protocol isolates a minimum viable cross-sectional ranker before adding retrieval, debate, tool use, or portfolio optimization layers. Negative results are equally informative: GPT-4o mini persistently ranks the cross-section in the wrong direction, producing negative IC and anti-monotonic quintiles rather than merely noisy alpha. For benchmark design, rank IC, portfolio spreads, and style diagnostics should therefore be reported together: IC tests ordering quality, spreads test economic investability, and factor diagnostics test whether apparent alpha is just familiar style compensation.

6 CONCLUSION

We studied LLMs as implicit factor-like priors for cross-sectional stock ranking, bridging LLM capability evaluation and factor diagnostics. The central lesson is that model choice is an asset-pricing choice: switching the LLM changes not only ranking accuracy, but also the growth, momentum, value, profitability, and investment exposures being tested. Across 11 included models, expected-return prompting is the primary lens: o3 reaches a 466.0% net long–short CAGR, Sharpe 5.54, and IC 0.594 after charging both long- and short-leg turnover, while GPT-4o and GPT-4o mini produce weak or negative signals. The input ablation shows that market summaries carry most of the pricing signal and factor-profile coherence, whereas fundamentals-only prompts weaken both alpha and style structure. For finance benchmarks, the implication is to report ordering quality, investable spread performance, and factor diagnostics together rather than treating an LLM ranking as a black-box forecast. Future work should extend to broader rolling universes, live model refreshes, and disclosure-text information axes while preserving the same audit trail from prompt to ranking, portfolio, and factor exposure.

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